

Problem 1. Lines in $\mathbb{R}^3$.

- (a) Find a parametrization for the line passing through $P = (-1, 0, 1)$ and $Q = (1, 2, 1)$.
- (b) Find a parametrization for the line of intersection of the following two planes:

$$\begin{cases} x + 2y + 2z = 1, \\ x + 3y + 5z = 3. \end{cases}$$

Problem 2. Integration of Vector-Valued Functions. Consider a parametrized curve $\mathbf{r}(t) = \langle x(t), y(t) \rangle$ with the following properties:

$$\mathbf{r}(0) = \langle 2, 1 \rangle,$$

$$\mathbf{r}'(0) = \langle 0, 1 \rangle,$$

$$\mathbf{r}''(t) = \langle -\cos t, -\sin t \rangle.$$

- (a) Integrate $\mathbf{r}''(t)$ to obtain $\mathbf{r}'(t)$.
- (b) Integrate $\mathbf{r}'(t)$ to obtain $\mathbf{r}(t)$.